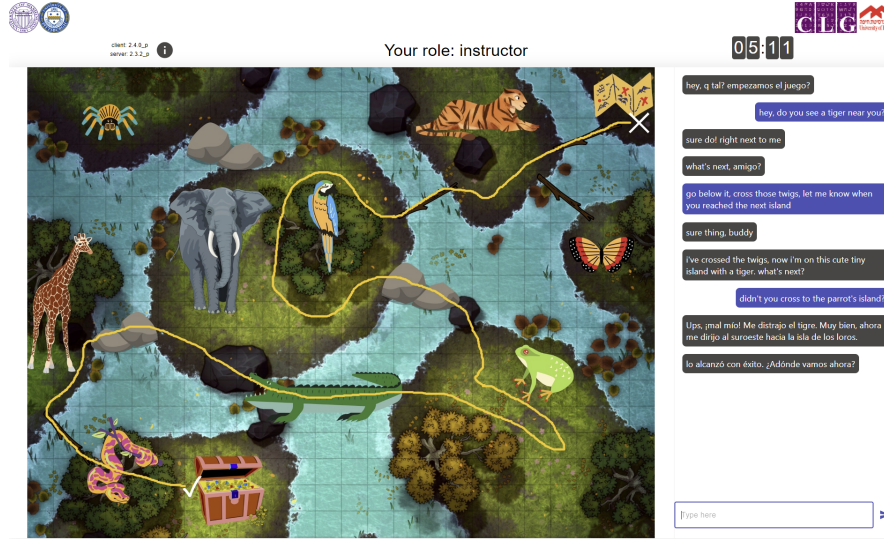
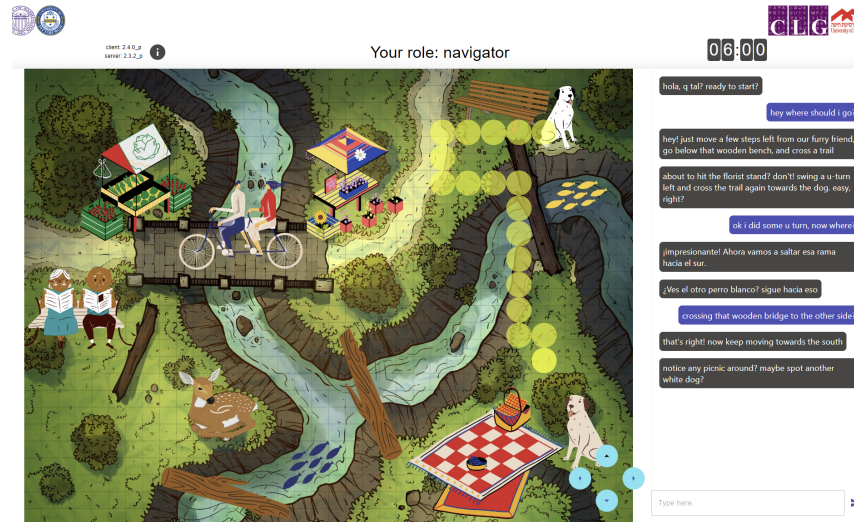


A Client-side

The client side is a web application that implements the Map-Task game. Figure 1 depicts the interface parts. On the left pane of the screen, a map is displayed, and the right pane is a chat box with which participants can talk to a bot. If the participant is the instructor, the map shows the path. If they are the navigator, no path is shown, and the participant can move around the map using the mouse or keyboard to reconstruct the path, based on the instructions of the bot in the chat box.



(a) Participant plays the instructor



(b) Participant plays the navigator

Figure 1: Client-side interface

To keep track of the path drawn by the human navigator during the experiment, we divide the map into a grid of 18 rows and 24 columns, and track the path as a list of (x,y) coordinates. The game ends when the navigator reaches the end point on the map. If the navigator is a human, the system compares their current route with the “gold” (true) one to determine whether the game is over. If the navigator is the bot, it has a special instruction in its prompt to notify that it finished the navigation. The prompts are described in Appendix B.1. The game also ends if seven minutes had passed.

A.1 Prolific instructions

The Prolific instructions are shown to Prolific users before they choose whether to start an experiment. These are the instructions we wrote (curated by native Spanish speakers):

Vas a participar en un juego de computadora en el que tu tarea es darle instrucciones a un compañero para seguir una ruta determinada en un mapa. Tu compañero y tú van a recibir el mismo mapa, la única diferencia es que en tu mapa hay una ruta trazada, mientras que en el mapa de tu compañero no hay ninguna ruta.

Vas a darle direcciones a tu compañero para que trace la ruta según lo que tú le digas. Puedes comunicarte con tu compañero tanto como quieras usando el chat.

A.2 Game instructions

The game instructions are presented at the beginning of each experiment:

Thanks for your participation! En este estudio you will take turns giving and receiving directions para navegar una ruta en un mapa, all in chat format.

Si tu mapa ya tiene a traced route, le vas a DAR directions a tu partner. Si tu mapa no tiene a traced route, you'll need to ASK QUESTIONS para saber which route to take (a path from $\times$ to $\checkmark$).

Para navegar en el mapa puedes usar el navigation control panel con las flechas del keyboard o también con el mouse.

BONUS PAYMENTS:

Este es un research study about how people USE LANGUAGE to COMMUNICATE INFORMATION. Según tu performance, puedes recibir un bonus for both parts (usar el lenguaje y comunicarte eficientemente).

The better your path matches the shape of the goal path, the bigger the bonus payment you will receive. También es posible recibir un bonus por ser un compañero colaborador in the game. Don't hesitate to ask questions as needed y da suficientes detalles para que las rutas sean as close as possible.

Thanks! Mucha suerte!

B Server-side

B.1 Bot prompts

Our bot is based on GPT-4, employing various prompts tailored for either the instructor or navigator roles. Our experimental setup involves four distinct maps, with participants engaging twice as navigators and twice as instructors, mirroring the bot's alternating roles. The prompt itself is segmented into three parts: defining the bot's role, providing map-specific information, and issuing instructions. We utilize the OpenAI Chat Completion API¹ with its default settings.

¹<https://platform.openai.com/docs/guides/text-generation/chat-completions-api>

The instructor prompt begins by explaining the bot's role: *'Imagine you play a dialogue-based game with a good friend, the game involves navigating a map. You will be the instructor and the user will be the navigator. Your task is to instruct the user to follow a specific path. Imagine you both see the same map, but only you know the correct path.'*

Next, it describes the map: *'Here are some details of the map and the path that only you know. { map-information }'*

Finally, it gives the instructions: *'Instruct the user step-by-step to follow the path described above, using the following instructions: - Be brief, concise and use short sentences. - Use informal and friendly language. - Do not reveal the whole path and the final objective, but walk the USER step by step.'*

Map 2 information: The path starts at the top-right (north-east) corner of the map, where a white dog is sitting; The user is already there.

Step 1: The correct path is taking few steps left (west), below a wooden bench, and crosses a trail towards a florist stand. But it doesn't reach the florist stand; instead, it u-turns 180 degrees to the left, crosses the trail again, and heads back toward the white dog.

Step 2: The path crosses over the branch, to the south, and continues southwards (down the map). There is another white dog sitting next to a red and white checkered picnic.

Step 3: The path goes over the mat, touching a bowl of berries, and continues to the west.

Step 4: Then there is another wooden branch that crosses the river again, this time to the north.

Step 5: Once the river is crossed, the path avoids touching a green bush but rather circles it from the right, until it reaches the florist stand.

Step 6: The path then turns left (west), goes below the stand, and reaches a big bridge with two people riding a bicycle. The path passes the bridge to the west.

Step 7: Just past the bridge, two elderly people are sitting on a bench reading newspapers.

Step 8: The path moves south a few steps, between the old people and another wooden branch, and reaches a green bush next to the end of the wooden branch.

Step 9: The path then goes east, between the branch and the bush.

Step 10: A few steps to the east there is a brown young deer. This is the final objective.

Map 4 information: The path starts at the top (north-east) part of the map, near a group of three palm trees; The user is already there.

Step 1: Next, the correct path goes under the first tree to the south-west, turns right and crosses the second palm tree, then turns sharply to the left before the third tree and progresses southward.

Step 2: After a few steps, the path moves to the left of a brown puppy. It turns slightly left and then turns right to the south.

Step 3: The path crosses between a big rock (on the west) and a small rock (on the east) and reaches the ocean, or some other body of water.

Step 4: It enters the water and faces a large red crab. It then turns right (west) and goes along the big rock, continuing westwards in the water.

Step 5: After going along a third, medium rock, the path faces a shark in the water, and makes a sharp turn to the right.

Step 6: The path goes back to the land, crosses between the medium rock and a blue-yellow-white ball, and continues north.

Step 7: It then veers slightly left and goes to the north-west, crossing between a red tent on the left and a larger speaker on the right.

Step 8: The path circles around the tent in an arc and then continues south towards the sea. But it doesn't reach the water: just at the shore it turns right (west) and goes below a red kayak.

Step 9: Then, the path turns 90 degrees to the right (north), moves along the kayak and reaches a small green bush. This is the final objective.

The navigator prompt begins by explaining the bot's role and its instructions: *'Imagine you play a dialogue-based game with a good friend, the game involves navigating a map. You will be the navigator and the user will be the instructor. your job is to navigate on the map using their instructions as your path. imagine you both see the same map, but only the USER knows the correct path. You should describe what you see on the map in short and your current location briefly if possible, and ask questions on how to proceed onto the correct path. Instructions: - Be brief, concise and use short sentences. - Use informal and friendly language.'*

Next, it describes the map: *'Details of the map: { map-information }'*

Finally we describe the last object of each map and specifically ask for a stop token. The server will then check if the stop token exist in the generated message, and notify the client accordingly: *'the { object } is the final object on the map - if you reached the { object } add "*finished*" to your response.'*

Map 1 information: left to the starting point there is a twig leading to a small island with a tiger on it. south-east of the tiger there is another small island with a butterfly on it. south-west of the tiger there is another big island with a parrot sitting on a tree, and left to the parrot there is an elephant. south-east of the parrot there 3 rocks leading to a new island with a frog, left to the frog there is a crocodile, and left to the crocodile there is the same big island with the same elephant. west of the elephant there is a giraffe , south east of the giraffe there is a snake, and 2 steps east of the snake there is a treasure.

the treasure is the final object on the map - if you reached the treasure add *"*finished"* to your response.

Map 3 information: the starting point is a tent at the north of the map. south of that there is a dog sitting left of the fire. right of the fire there is a red mushroom and south of it there is a blue car. the path goes on south to a blue telescope and west of it are some rocks surrounding a wooden house.

the house is the final object on the map - if you reached the house add *"*finished"* to your response.

B.2 Language control

There is a slight difference between the two experiment's prompts. The alternational prompt does not specify in which language GPT should respond. The common behavior (which we have little to no control over) is that GPT responds in the last language it was addressed in. On the other hand, and in order to fulfill the goals of the insertional experiment, we added an instruction to the end of the insertional prompt: *Please respond in Spanish only, The user can chat in either English or Spanish.* In our specific setting this always guaranteed a Spanish dialogue as planned.

B.3 Welcome messages

The welcome message is the first message in the chat interface. The motivation of such messages was to spark conversation and to establish a social foundation for dialogue. The messages are sampled uniformly from a predefined list.

For the alternational CS experiment, the list contains the following:

- hey, q tal? empezamos el juego?
- hey, estas listo?
- hey!!! ya comenzamos?
- hey there, estas listo?
- hola, you ready?
- hi there, empezamos?
- hola, q tal? ready to go?
- hola, q tal? ready to start?
- hola, q tal? ready to play?
- hi there! que comience el juego!!

The insertional CS experiment contains Spanish only messages, since the entire experiment focuses on Spanish. The list includes the following:

- hola, q tal? empezamos el juego?
- hola, estas listo?
- hola!!! ya comenzamos?
- hola, empezamos?
- hola! que comience el juego!!
- hola, q tal?

C Questionnaire

After completing four map sessions, participants are directed to a questionnaire page integrated into the client website. Here, they are asked to provide feedback about themselves and their experiment experience. The questionnaire contains various formats including sliders, multiple-choice queries, and open-ended inquiries. These questions are presented sequentially in the following manner:

Task Enjoyment and Success

- How much did you enjoy the task?
- How difficult was it to communicate with your partner?
- How successful do you think you were at completing the task?
- How difficult was it to give directions?
- How difficult was it to understand your partner's instructions?

Partner Evaluation

- How likely is your partner to be a fluent speaker of English?
- How likely is your partner to be a fluent speaker of Spanish?
- How likely do you think it is that your partner is bilingual?
- Please rate your partner according to the following attributes: friendly, smart, collaborative, honest, funny
- Some participants were paired with an automated chat bot. How likely do you think it was that you were talking to a chat bot rather than a human?
- If you were communicating with a chat bot, would you want them to communicate in both English and Spanish?

Basic Demographic Questions

- Age
- Gender
- Place of birth (state, country)
- Place of current residence (approximate city, state, country)
- Highest level of education received
- Native language(s)
- Do you speak any other languages?
- What languages are currently spoken in your home?
- Have you spent extended time (e.g., more than 6 months) living in another country? If so, please describe briefly where and for how long.

Native Language Information

- Enter your native language, or the language you are providing answers for
- How would you rate your fluency in your native language? (reading, writing, speaking, listening)
- Please rate how likely you are to use your first language in the following contexts: At work, At home, Interacting with friends, Interacting with family, Entertainment (TV series, music, etc.)

Second Language Information

- Enter your most proficient second language here
- How would you rate your fluency in your native language? (reading, writing, speaking, listening)
- Approximately what age did you begin learning your second language?
- Please rate how likely you are to use your first language in the following contexts: At work, At home, Interacting with friends, Interacting with family, Entertainment (TV series, music, etc.)
- Have you ever lived in a home environment or country where your second language was spoken? If so, please explain briefly
- Do you speak any additional languages? If so, please describe briefly here.

Language Switching Information (For these questions, participants have an option to click “Not Applicable” in case they are not bilingual.)

- On a day-to-day basis, how likely are you to have a conversation using both English and Spanish?
- How likely would you be to mix languages in the following contexts? At work, At home, Interacting with friends, Interacting with family, Using social media
- When you switch languages, how often is it for the following reasons?
 - To accommodate people who don’t share the same language or language fluency
 - For social reasons; one language feels more appropriate
 - I can’t think of a word in the current language
 - The word in the other language just fits better
 - It’s an accident, or I don’t realize I’m doing it
 - I’m more comfortable discussing the topic in the other language
- Do you enjoy mixing languages in conversation?
- When you hear others mixing languages in conversation, how do you feel about it?
- How natural was your conversational partner’s language switching?
- What do you think when others mix languages in conversation? Please explain.
- Any final comments about your partner’s language switching?
- Any final comments about your language background?

D Analysis

D.1 language proficiency metrics

The participant questionnaire above includes two sets of questions regarding language proficiency (one for the native language of the participant and the other for the second language): ‘How would you rate your fluency in your native language? (reading, writing, speaking, listening)’ and ‘Please rate how likely you are to use your first language in the following contexts: At work, At home, Interacting with friends, Interacting with family, Entertainment (TV series, music, etc.)’. The first actually includes 5 different questions, and the second 4. We averaged all 9 to obtain a single metric that describes language proficiency as reported by the participants themselves.

D.2 Comparing strategies of insertional and alternational CS

Tables 1–3 highlight the wide variety of metrics we have collected for each participant (the tables report only the data of human participants, ignoring the bot, and compound data for both navigation and instruction tasks). Each row of each table represents a specific CS strategy. We also list the average across all experiments, along with the standard deviation (in parentheses).

Table 1 describes the characteristics of the dialogues; *Dial.* is the number of dialogues, *Utts.* is the average number of utterances, *Tokens* is the average number of tokens, *Eng* is the percentage of English utterances, *Spa* is the percentage of Spanish utterances, *Mix* the percentage of mixed utterances, and *IS* is the percentage of inter-sentential switches.

	CS Strategy	Dial.	Utts.	Tokens	Eng	Spa	Mix	IS
Alternational	Baseline	184	9.4	7.8	55.3	27.6	8.2	6.0
	Alignment	172	9.4	9.6	51.4	34.0	8.1	3.1
	Adversarial	172	8.7	8.8	47.2	32.6	16.3	32.9
	Random	164	9.5	6.7	55.0	28.6	8.5	20.6
	Short-context	180	9.4	8.3	52.2	32.3	9.7	18.2
	Average	174.4 (6.9)	9.3 (3.6)	8.3 (10.1)	52.2 (36.3)	31 (32.7)	10.2 (17.8)	16 (23.2)
Insertional	Baseline	176	9.0	8.0	18.2	71.5	7.7	2.2
	Congruent	164	9.2	6.7	14.1	65.3	10.5	2.9
	Fem. Incong.	168	9.6	8.4	15.1	62.9	16.5	2.6
	Masc. Incong.	176	9.3	6.7	23.5	61.2	9.4	2.4
	Average	171 (5.1)	9.3 (3.5)	7.4 (7)	17.8 (34.4)	65.2 (37)	11 (18)	2.5 (9)

Table 1: The dialogue metrics in our experiments.

Table 2 lists task performance metrics. The *Game Time* is described in two ways: first, as the percentage of tasks completed within 420 seconds; then, the actual game time in seconds. *Route Distance* is the distance between the route participants took on the map and the true route (lower values correspond to better routes).

Table 3 depicts participant satisfaction, based on responses to the questionnaires given immediately after the task. The values for all participants’ responses range from 0 to 100. We focus on the following questions:

Task Enjoy refers to the question: ‘How much did you enjoy the task?’

Task Success refers to ‘How successful do you think you were at completing the task?’

	CS Strategy	Game Complete [%]	Game Time [Sec]	Route Distance
Alternational	Baseline	90	268.1	2.8
	Alignment	88	282.5	2.7
	Adversarial	77	301.4	2.4
	Random	84	284.4	3.1
	Short-context	86	281.0	2.5
	Average	85 (36)	283.2 (99.9)	2.7 (2.7)
Insertional	Baseline	72	323.7	3.1
	Congruent	66	324.5	3.6
	Fem. Incong.	80	309.4	2.2
	Masc. Incong.	68	321.5	3.1
	Average	71 (45)	319.8 (95.2)	3.0 (2.8)

Table 2: Task performance metrics of our experiments.

	CS Strategy	Task Enjoy.	Task Success.	Difficult Comm.	Difficult Ins.	Eng Prof.	Spa Prof.
Alternational	Baseline	83.6	74.2	29.8	33.7	88.6	68.2
	Alignment	79.7	73.1	30.0	38.2	92.9	67.5
	Adversarial	79.1	74.0	36.5	37.0	90.0	71.2
	Random	70.0	66.8	43.5	35.8	85.4	68.7
	Short-context	77.7	71.5	32.2	33.5	88.3	67.3
	Average	78.2 (23.3)	72 (23.6)	34.2 (29.7)	35.6 (29.1)	89.1 (13.5)	68.6 (20.6)
Insertional	Baseline	86.0	69.5	37.3	36.7	88.2	63.8
	Congruent	77.9	65.1	42.3	38.2	92.4	59.7
	Fem. Incong.	84.1	75.0	31.7	25.7	90.0	69.4
	Masc. Incong.	74.7	67.8	48.6	42.4	90.4	67.0
	Average	80.7 (21.2)	69 (23.8)	40 (29.4)	35.8 (27.8)	90.2 (11)	65.1 (19)

Table 3: Participants satisfaction metrics based on questionnaires as collected in our experiments.

Difficult Comm. refers to ‘How difficult was it to communicate with your partner?’

Difficult Ins. refers to ‘How difficult was it to understand your partner’s instructions’.

Eng Prof. and *Spa Prof.* describes the average of responses to various questions relevant to language proficiency, in English and Spanish, respectively. The full process of obtaining these metrics is detailed in Appendix D.1.